

	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Semester Examination – Summer 2022 Course: B. Tech. in Mechanical Engineering Subject Name: Quantitative Techniques in Project Management (QTPM) Subject Code: BTMEC605A Max Marks: 60 Date: Duration: 3 Hr.																	
	Instructions to the Students: 1. All the questions are compulsory. 2. The level question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.																	
					(Level/CO)	Marks												
Q.1	Solve the following.																	
A)	Maximize $Z = 80X_1 + 120X_2$ Subjected to $X_1 + X_2 \leq 9$ $20X_1 + 50X_2 \leq 360$ $X_1 \geq 2$ $X_2 \geq 3$ $X_1, X_2 \geq 0$					12												
	OR																	
A)	A company manufactures two products X and Y whose profit contributions are Rs.10 and Rs. 20 respectively. Product X requires 5 hours on machine I, 3 hours on machine II and 2 hours on machine III. The requirement of product Y is 3 hours on machine I, 6 hours on machine II and 5 hours on machine III. The available capacities for the planning period for machine I, II and III are 30, 36 and 20 hours respectively. Find the optimal product mix. Solve by Simplex method.					12												
Q.2	Solve any Two of the following.																	
A)	Explain the linear programming model of the transportation problems with suitable example. Also explain any one method of Transportation for obtaining the initial basic feasible solution.					6												
B)	An airline company has drawn up a new flight schedule involving five flights. To assist in allocating five pilots to the flights, it has asked them to state their preference scores by giving each flight a number, the greater is preference. Some of these flights are unsuitable to same pilots owing to domestic reasons. <table border="1"><tr><td>Flight No.</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>Pilot A</td><td>8</td><td>2</td><td>--</td><td>5</td><td>4</td></tr></table>					Flight No.	1	2	3	4	5	Pilot A	8	2	--	5	4	6
Flight No.	1	2	3	4	5													
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		<table><tr><td>Pilot B</td><td>10</td><td>9</td><td>2</td><td>8</td><td>4</td></tr><tr><td>Pilot C</td><td>5</td><td>4</td><td>9</td><td>6</td><td>--</td></tr><tr><td>Pilot D</td><td>3</td><td>6</td><td>2</td><td>8</td><td>7</td></tr><tr><td>Pilot E</td><td>5</td><td>6</td><td>10</td><td>4</td><td>3</td></tr></table>	Pilot B	10	9	2	8	4	Pilot C	5	4	9	6	--	Pilot D	3	6	2	8	7	Pilot E	5	6	10	4	3			
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	What would be the allocation of pilots to flights in order to meet as many preferences as possible.																												
C)	What do you mean by balancing an assignment problem? What steps you take to solve maximization case in assignment problem? Explain with an example.					Understand/CO 2	6																						
Q.3	Solve Any Two of the following																												
A)	A newspaper boy buys papers for 30 paise and sells them for 70 paise each. He can not return unsold newspapers. Daily demand has the following distribution. <table><tr><td>n</td><td>23</td><td>24</td><td>25</td><td>26</td><td>27</td><td>28</td><td>29</td><td>30</td><td>31</td><td>32</td></tr><tr><td>p</td><td>0.01</td><td>0.03</td><td>0.06</td><td>0.10</td><td>0.20</td><td>0.25</td><td>0.15</td><td>0.10</td><td>0.05</td><td>0.05</td></tr></table> If each days demand is independent of the previous days, how many papers should he orders every day?					n	23	24	25	26	27	28	29	30	31	32	p	0.01	0.03	0.06	0.10	0.20	0.25	0.15	0.10	0.05	0.05	Apply/CO3	6
n	23	24	25	26	27	28	29	30	31	32																			
p	0.01	0.03	0.06	0.10	0.20	0.25	0.15	0.10	0.05	0.05																			
B)	The demand for a product is 2400 units over 360 days. The storage cost is 0.06% of the unit cost of the product and the ordering cost is Rs. 35000. Find the optimal ordering quantity if the price breaks are as follows: Quantity range Purchasing cost (Rs.) 0 <q<1000 1000 1000<q<4000 925 4000<q 850					Apply/CO4	6																						
C)	Consider the inventory system with the following data in usual notations: R = 1000 units/year, I = 0.30, P = 0.50 per unit, C3 = Rs. 10, L= 2 Years, and C1 = I P, Calculate the Optimal order quantity, Reorder point and Minimum average cost.					Apply/CO4	6																						
Q.4	Solve any Two of the following.																												
A)	Derive the expression for Economic Lot Size (for manufacturing model) without shortage and deterministic Uniform demand.					Understand/CO4	6																						
B)	Explain with suitable examples about the queue. Why do you consider the study of waiting line as an important aspect? Explain the Kendall's notation used in queuing theory.					Remember/CO3	6																						
C)	The maintenance cost and resale value per year of a machine whose purchase price is Rs. 7000/-are given below: Year 1 2 3 4 Maintenance 900 1200 1600 2100 Cost (Rs.) Resale Value 4000 2000 1200 600 (Rs.) When should the machine be replaced?					Apply/CO5	6																						
Q.5	Solve any one of the following																												

